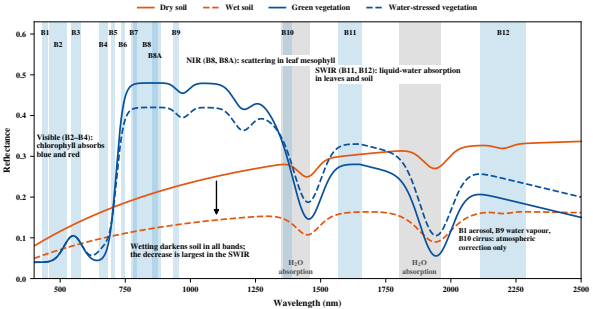
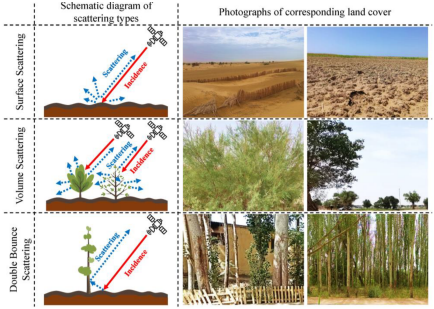
# Motivation Different ways of obtaining soil moisture

| Source | How SM is obtained | Depth | Main limitation |
|---|---|---|---|
| **Microwave** (SMAP, SMOS, ASCAT, S1) | invert: $\sigma^0$ or $T_B = f(\text{SM}, \text{roughness}, \text{vegetation})$ | top $\sim$5 cm; informs part of the root zone (Feldman et al., 2023) | shallow; assumes vegetation and roughness; coarse (9–40 km) except S1 |
| **Optical / thermal** (S2, Landsat, MODIS) | empirical relations with surface properties: reflectance and indices; LST–vegetation space (e.g. triangle method, TVDI) | skin / surface | indirect, site-specific calibration; clouds; canopy hides the soil |
| **Land-surface models** | $d\theta/dt = P - ET - Q - D$, with parameters $p$ | full profile | forcing errors, soil and vegetation parameters, process representation (Koster et al., 2009; Seneviratne et al., 2010) |
| **Reanalysis** (ERA5-Land) | LSM driven by reanalysis meteorology | full profile | model assumptions dominate deep layers with few observational constraints (Li et al., 2021; Miguez-Macho and Fan, 2021) |
| **In situ** (ISMN) | direct probe measurement | several depths | point scale; sparse, uneven global coverage |
| **Data-driven** (ML / DL) | SM $= f_\theta(\text{inputs})$, learned from in-situ labels; no ad-hoc process assumptions | any depth with labels | needs enough, representative sites; generalisation?; hard to interpret? |

# Motivation Multimodal remote sensing



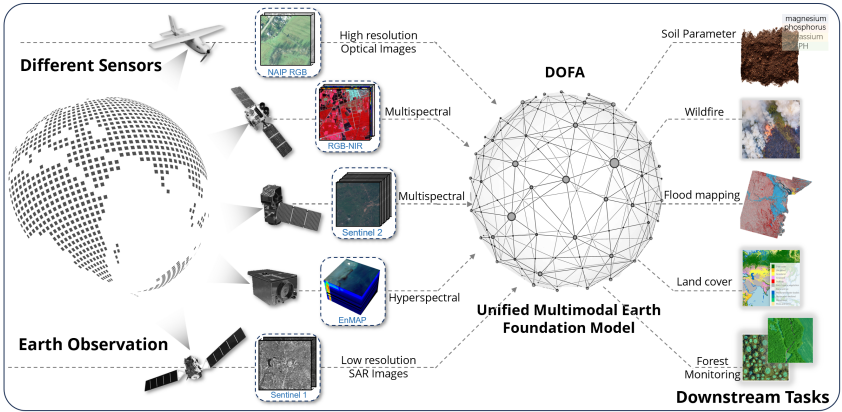

*Optical: dry vs wet soil; green vs water-stressed vegetation*




*SAR scattering types (Nurmemet et al., 2025)*

**Soil and vegetation water change both optical reflectance and SAR backscatter**

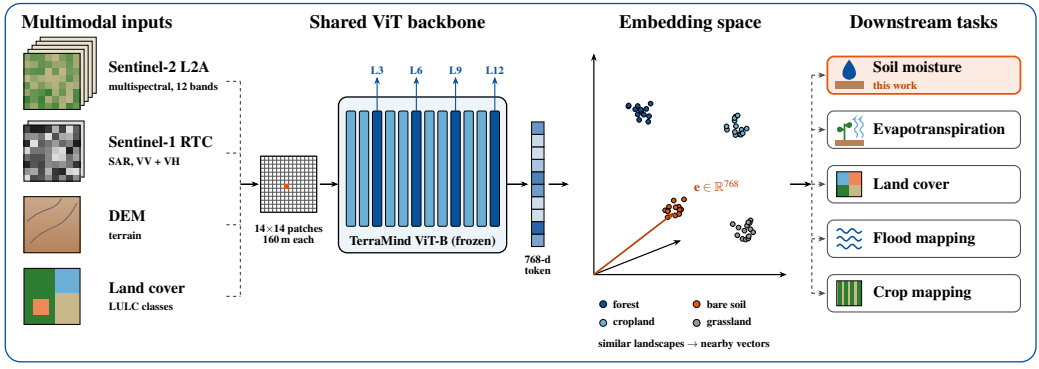
# Motivation Geospatial foundation models (GFMs)




*DOFA: one model for many sensors and many downstream tasks (Xiong et al., 2024, arXiv:2403.15356)*

**Once pre-trained on multimodal, unlabelled data, it works for many downstream tasks**

# Motivation TerraMind embeddings for soil moisture

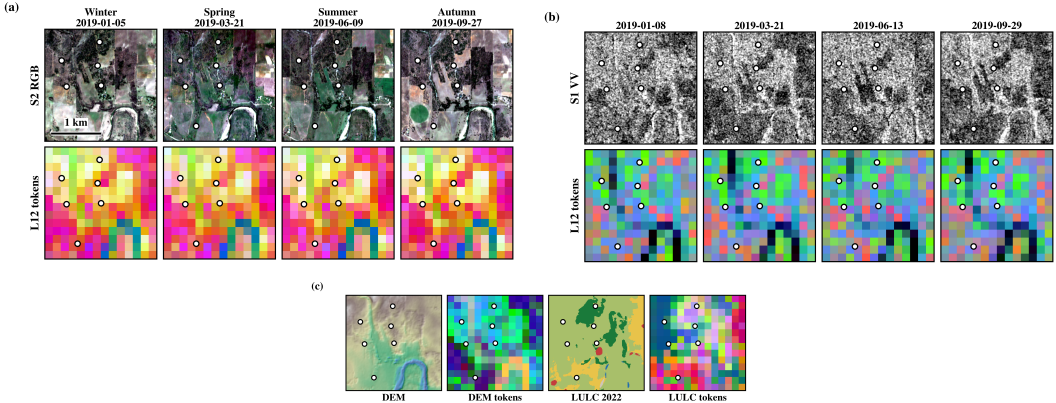



*S2, S1, DEM and LULC → frozen TerraMind ViT-B → one 768-d embedding per 160 m patch → downstream tasks*

**The same embedding can serve many tasks; in this case, soil moisture**

# Motivation

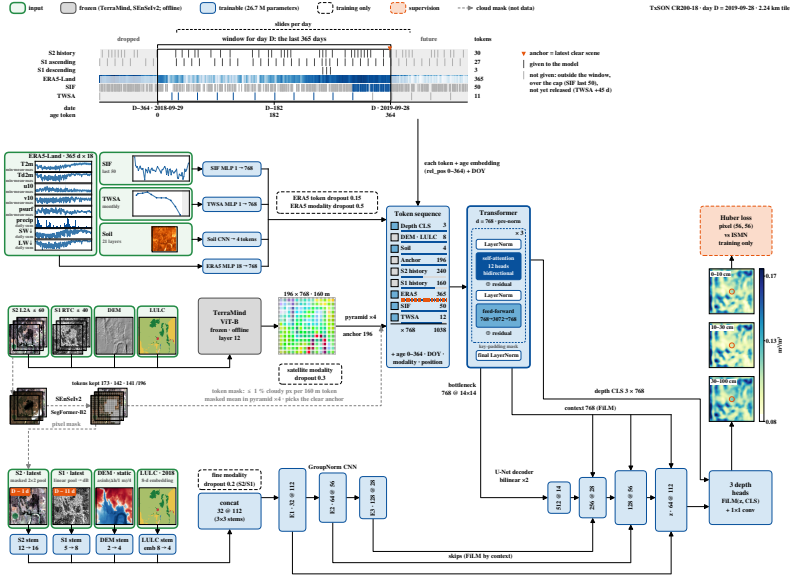
## Example: TerraMind embeddings at one site (TxSON, 2019)



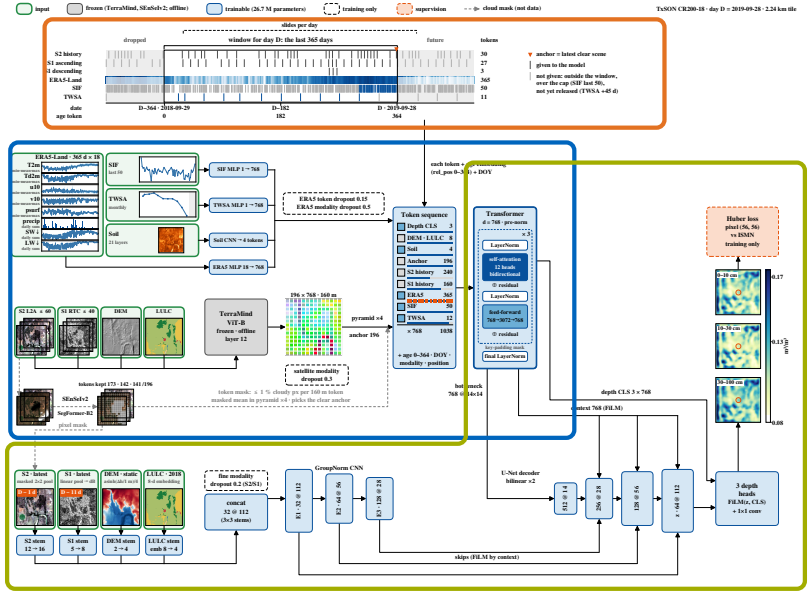

*(a) S2 in four seasons, (b) S1 VV, (c) DEM and LULC, each with its TerraMind L12 tokens*

**Each 160 m patch → one 768-d vector; colours show its first 3 principal components**

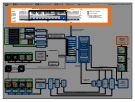
# Method Model architecture

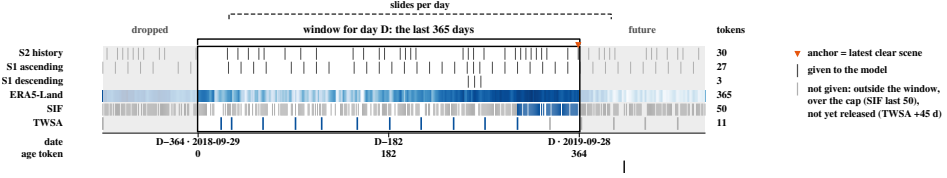


# Method Model architecture

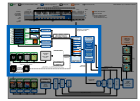


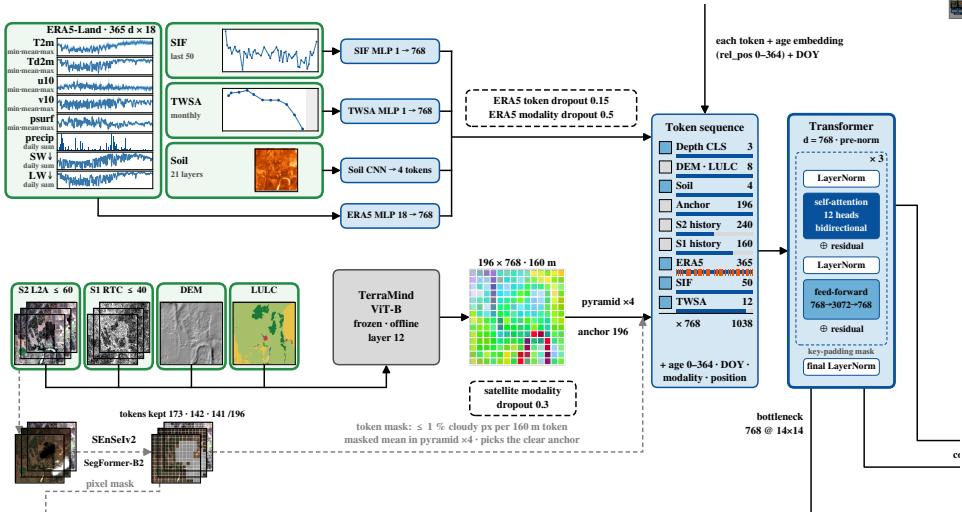
# Method Inputs: what the model sees for day D



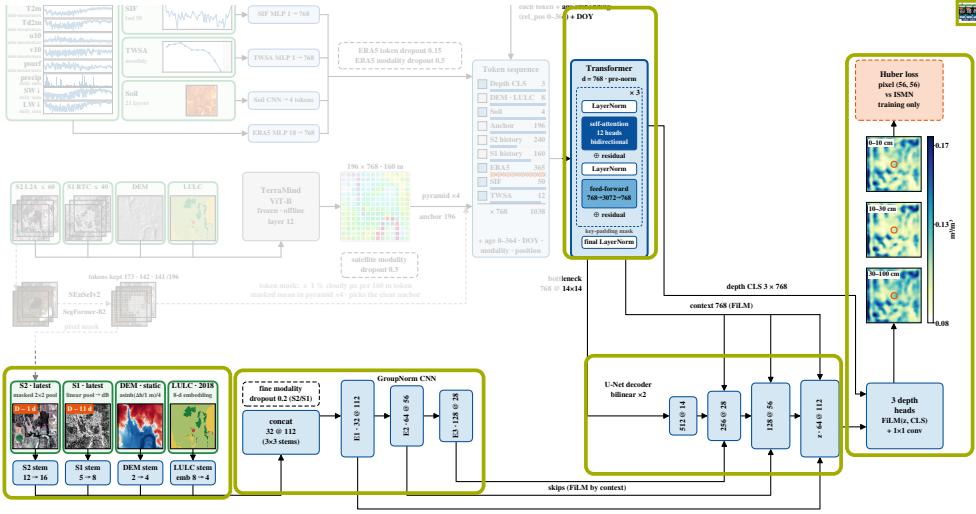


# Method Coarse path: tokens → transformer

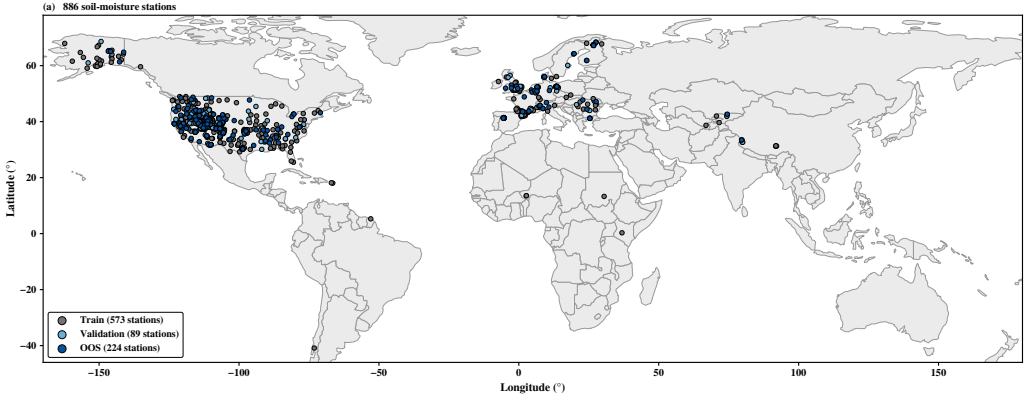




**Method** **Fine path: 20 m CNN → U-Net decoder → depth heads**



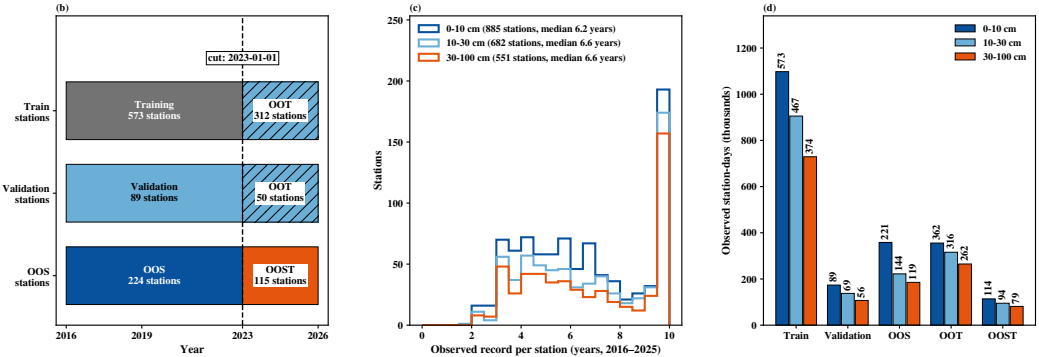
# Data In-situ soil moisture: global coverage




*886 soil-moisture stations (ISMN and flux sites): train 573, validation 89, out-of-sample (OOS) 224*

**Dense in North America and Europe; sparse in the tropics and the Southern Hemisphere**

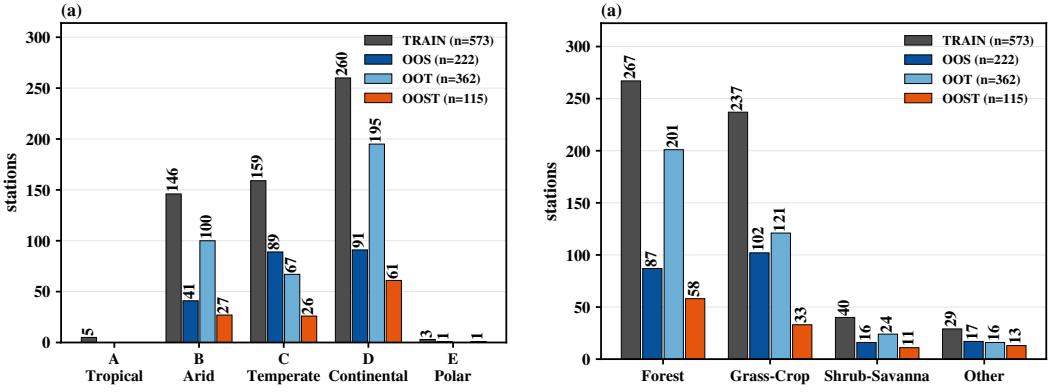
# Data Depth coverage, record length and splits




*(b) train / validation / OOS stations, time cut 2023-01-01; (c) observed record per station; (d) observed station-days*

**0–10 cm: 885 stations (median 6.2 yr); 10–30 cm: 682 (6.6 yr); 30–100 cm: 551 (6.6 yr)**

# Data Climate and land-cover coverage



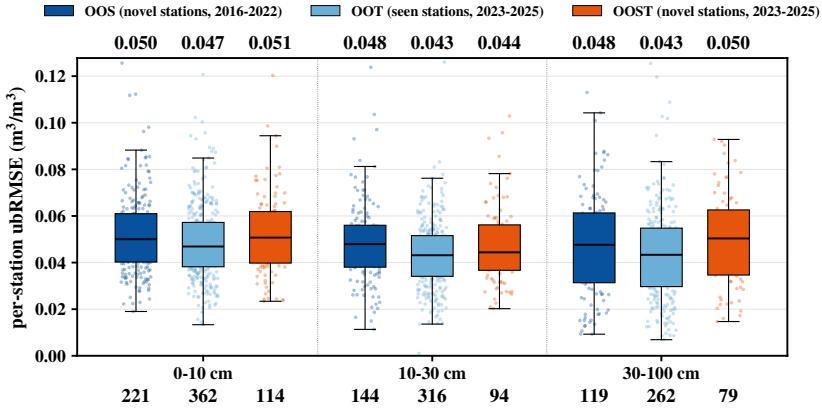

*Stations per Köppen climate zone (left) and land-cover class (right), by split*

**Mostly continental, temperate and arid; forest and grass–crop; almost no tropics or polar**

| | |
|---|---|
| **Stations** | 990 in inventory; **886** with soil moisture (train 573 / val 89 / OOS 224) |
| **Depths** | 0–10 cm: 885 10–30 cm: 682 30–100 cm: 551 stations |
| **Period** | 2016–2025; out-of-time test from 2023-01-01 |
| **Soil-moisture observations** | **5.14 M** observed daily values (2.10 M / 1.68 M / 1.37 M per depth) |
| **Sentinel-2 L2A** | 272,281 scenes (994 sites); cloud $< 75\,\%$ per scene, cloudy tokens masked |
| **Sentinel-1 RTC** | 496,835 scenes: 251,310 ascending + 245,525 descending |
| **ERA5-Land** | 18 variables, 2.46 M station-days (990 stations) |
| **Static / other** | DEM, LULC, soil (OpenLandMap), SIF, TWSA |

**About 5.1 million daily soil-moisture observations at 886 stations, 0–100 cm**

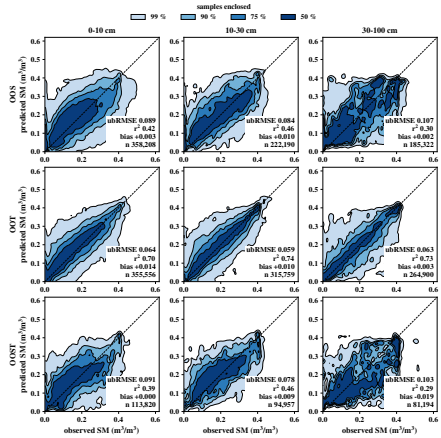
## Results Per-station ubRMSE by depth




*One point per station; medians above, station counts below*

**Median per-station ubRMSE 0.043–0.051 $m^3/m^3$ at every depth and split**

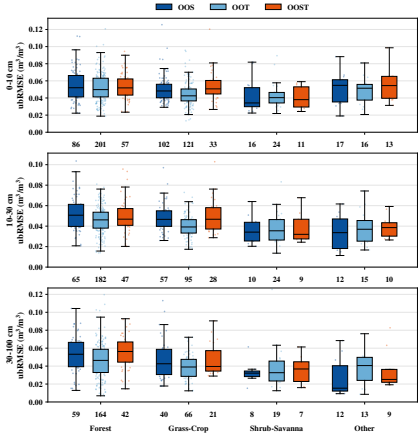
# Results Predicted vs observed, by split and depth




*All daily samples, smoothed density; shaded bands hold 50/75/90/99 % of samples; rows OOS, OOT, OOST*

**New years at known sites (OOT, $r^2$ 0.70–0.74) fit far better than new sites ($r^2$ 0.29–0.46)**

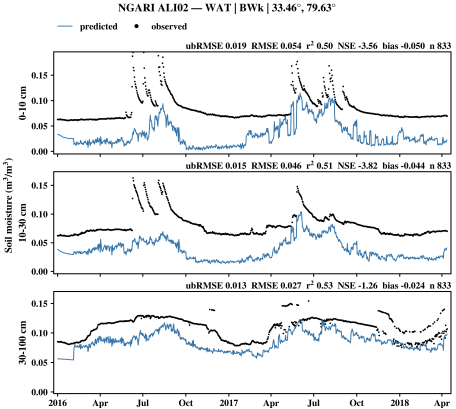
# Results Per-station ubRMSE by land cover



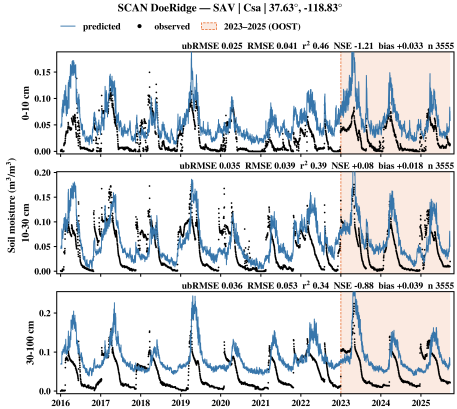

*Rows: depths; groups: land-cover class; colours: OOS, OOT, OOST*

**Similar error across land cover; forest slightly highest, shrub–savanna lowest**

# Results Sites where all three depths match




**NGARI ALI02, Tibetan Plateau (best)**

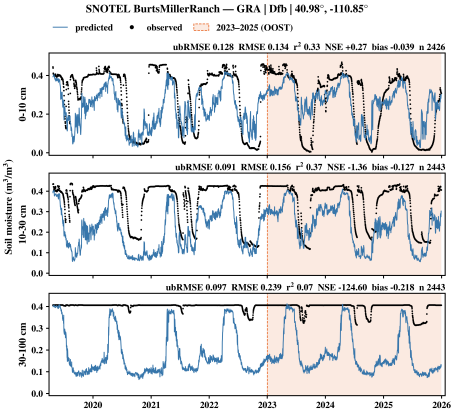



**SCAN Doe Ridge, California**

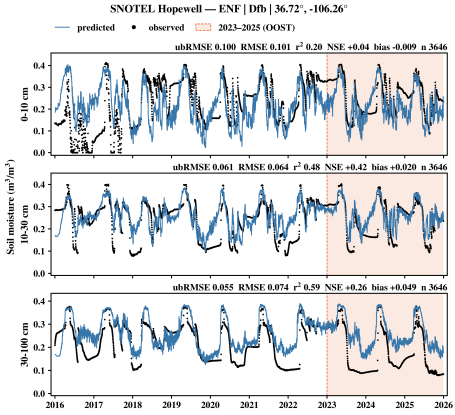
*OOS stations observed at all three depths, lowest ubRMSE; blue predicted, black observed*

**Dynamics right at all depths (ubRMSE 0.013–0.036, r 0.57–0.73); level off by up to 0.05**

# Results Sites where the depths do not match




**SNOTEL Burts Miller Ranch, Utah**

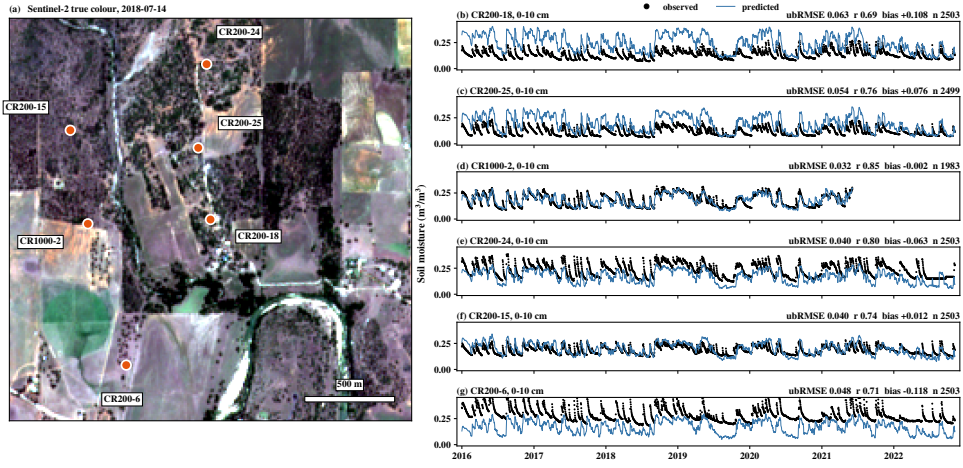



**SNOTEL Hopewell, New Mexico**

*OOS stations observed at all three depths, highest ubRMSE; shaded = 2023–2025 (OOST)*

**Snowmelt peaks timed right; summer dry-down wrong: too dry deep at Burts Miller, too wet at Hopewell**

## Results TxSON: six stations inside one 2.2 km tile




*(a) Sentinel-2 true colour with the six stations; (b–g) 0–10 cm, predicted vs observed*

**Timing right at every station (r 0.69–0.85); offsets remain (bias −0.12 to +0.11)**